Mathematical Foundations of Political Methodology

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Day 7: CLT

- Moment Generating Functions
- Limit Theorems
- Properties of Estimators

Approximating functions and second order conditions

Theorem

Taylor's Theorem *Suppose $f : \mathbb{R} \to \mathbb{R}$, $f(x)$ is infinitely differentiable function. Then, the Taylor expansion of $f(x)$ around a is given by*

$$
\begin{aligned}
f(x) &= f(a) + \frac{f^{'}(a)}{1!}(x-a) + \frac{f^{''}(a)}{2!}(x-a)^2 + \frac{f^{'''}(a)}{3!}(x-a)^3 + \ldots \\
f(x) &= \sum_{n=0}^{\infty} \frac{f^{n}(a)}{n!}(x-a)^n
\end{aligned}
$$

Example Function

Suppose $a = 0$ and $f(x) = e^x$. Then,

$$\begin{array}{rcl} f^{'}(x) & = & e^x \\ f^{''}(x) & = & e^x \\ \vdots & \vdots & \vdots \\ f^{n}(x) & = & e^x \end{array}$$

This implies

$$e^x \quad = \quad 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \ldots + \frac{x^n}{n!} + \ldots$$

Moment Generating Functions

Definition

Suppose X is a random variable with pdf f. Define,

$$E[X^n] \;=\; \int_{-\infty}^{\infty} x^n f(x)dx$$

We will call X^n the n^{th} moment of X

- By this definition $var(X) = \text{Second Moment} - \text{First Moment}^2$
- We are assuming that the integral converges

Moment Generating Functions

Proposition

Suppose X is a random variable with pdf $f(x)$. Call $M(t) = E[e^{tX}]$,

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \end{aligned}$$

We will call $M(t)$ the moment generating function, because:

$$\frac{\partial^n M(t)}{\partial^n t}\Big|_0 = E[X^n]$$

(Assuming that we can interchange derivative and integral)

Moment Generating Functions

Proof.

Recall the Taylor Expansion of e^{tX} at 0,

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Then,

$$E[e^{tX}] = 1 + tE[X] + \frac{t^2}{2!}E[X^2] + \frac{t^3}{3!}E[X^3] + \dots$$

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Differentiate once:

$$\begin{aligned}\frac{\partial M(t)}{\partial t} &= 0 + E[X] + \frac{2t}{2!}E[X^2] + \dots \\ M^{'}(0) &= 0 + E[X] + 0 + 0\dots\end{aligned}$$

□

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$$\frac{\partial^n M(t)}{\partial^n t} = 0 + 0 + 0 + \ldots + \frac{n \times n - 1 \times \ldots 2 \times t^0 E[X^n]}{n!} + \frac{n! t E[X^{n+1}]}{(n+1)!} + \ldots$$

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- If two random variables, X and Y have the same moment generating functions, then $F_X(x) = F_Y(y)$ for almost all x.

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\end{aligned}$$

Proposition

Suppose X_i are a sequence of independent random variables. Define

$$Y = \sum_{i=1}^{N} X_i$$

Then

$$M_Y(t) = \prod_{i=1}^{N} M_{X_i}(t)$$

Proof.

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□

Sequences and Convergence

Sequence (refresher):

$$\{a_i\}_{i=1}^{\infty} \quad = \quad \{a_1, a_2, a_3, \ldots, a_n, \ldots,\}$$

Definition

We say that the sequence $\{a_i\}_{i=1}^{\infty}$ converges to real number A if for each $\epsilon > 0$ there is a positive integer N such that for $n \geq N$, $|a_n - A| < \epsilon$

Sequences and Convergence

Sequence of functions:

$$\{f_i\}_{i=1}^{\infty} \quad = \quad \{f_1, f_2, f_3, \ldots, f_n, \ldots, \}$$

Definition

Suppose $f_i : X \rightarrow \Re$ for all i. Then $\{f_i\}_{i=1}^{\infty}$ converges pointwise to f if, for all $x \in X$ and $\epsilon > 0$, there is an N such that for all $n \geq N$,

$$|f_n(x) - f(x)| < \epsilon$$

This is as strong of a statement as we're likely to make in statistics

Convergence Definitions

Define $\widehat{\theta}_n$ to be estimator for θ based on n observations.

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- What is the probability $\widehat{\theta}_n$ differs from θ?
- What is the probability $\left\{\widehat{\theta}_i\right\}_{i=1}^{n}$ converges to θ?
- What is sampling distribution of $\widehat{\theta}_n$ as $n \rightarrow \infty$?

Convergence in Probability

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Definition

We will say the sequence $\widehat{\theta}_n$ *converges in probability to* θ *(perhaps a non-degenerate RV) if,*

$$\lim_{n \to \infty} Prob(|\widehat{\theta}_n - \theta| > \epsilon) = 0$$

For any $\epsilon > 0$

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- ϵ is a tolerance parameter: how much error around θ?
- In the limit, convergence in probability implies sampling distribution collapses on a spike at θ
- $\left\{\widehat{\theta}_i\right\}$ need not actually converge to θ, only P($|\theta_n$ - $\theta| > \epsilon) = 0$

Example (Cassella and Burger)

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$$P(|X_n - X| > \epsilon) \quad = \quad P(s \in [\mathtt{l}_n, \mathtt{u}_n])$$

Length of $[\mathtt{l}_n, \mathtt{u}_n] \rightarrow 0 \Rightarrow P(s \in [L_n, U_n]) = 0$

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$$\widehat{\theta}_n(s) \quad \rightarrow \quad \theta(s)$$

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Definition

We will say the sequence $\widehat{\theta}_n$ converges almost surely to θ if,

$$Prob(\lim_{n\to\infty} |\widehat{\theta}_n - \theta| > \epsilon) = 0$$

- This is stronger than convergence in probablity, it says that sequence converges to θ (almost everywhere))
- Think about definition of random variable: $\widehat{\theta}_n$ is a function from sample space to real line.
- Almost sure says that, for all outcomes (s) in sample space (S) $s \in S$,

$$\widehat{\theta}_n(s) \quad \rightarrow \quad \theta(s)$$

Except for a subset $\mathcal{N} \subset S$ such that $P(\mathcal{N}) = 0$.

Example (Cassella and Berger)

Suppose $S \sim$ Uniform(0,1).
Suppose X_n is define as follows:

$$\begin{aligned} X_1(s) = s + I(s \in [0,1]) \quad &, \quad X_2(s) = s + I(s \in [0,1/2]) \\ X_3(s) = s + I(s \in [1/2,1]) \quad &, \quad X_4(s) = s + I(s \in [0,1/3]) \\ X_5(s) = s + I(s \in [1/3,2/3]) \quad &, \quad X_6(s) = s + I(s \in [2/3,1]) \end{aligned}$$

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Does $X_n(s)$ converge almost surely to $X(s) = s$?
No!: the sequence doesn't converge for each s
For each value of s the sequence varies between s and $s+1$ infinitely often

Convergence in Distribution

We've talked about $\widehat{\theta}_n$'s sampling distribution converging to a normal distribution.

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A series of random variables, $X_1,\ X_2,\ \ldots$ with cdf $F_n(x)$ converges in distribution to random variable Y with cdf $F(x)$ if

$$\lim_{n\to\infty} F_n(x) - F(x) = 0$$

For all $x \in \Re$ where $F(x)$ is continuous.

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- Weakest form of convergence almost sure $\rightarrow$ probability $\rightarrow$ distribution
- Says that cdfs are equal, says nothing about convergence of underlying RV
- Useful for justifying use of some sampling distributions

Convergence in Distribution $\not\Rightarrow$ Convergence in Probability

Define $X \sim N(0,1)$ and each $X_n = -X$. Then:
$X_n \sim N(0,1)$ for all n so X_n trivially converges to X. But,

$$\begin{aligned} P(|X_n - X| > \epsilon) &= P(|X + X| > \epsilon) \\ &= P(|2X| > \epsilon) \\ &= P(|X| > \epsilon/2) \not\rightsquigarrow 0 \end{aligned}$$

Central Limit Theorem

Proposition

Let X_1, $X_2, \ldots$ be a sequence of independent random variables with mean μ and variance σ^2. Let X_i have a cdf $P(X_i \leq x) = F(x)$ and moment generating function $M(t) = E[e^{tX_i}]$. Let $S_n = \sum_{i=1}^{n} X_i$. Then

$$\lim_{n\to\infty} P\left(\frac{S_n - \mu n}{\sigma\sqrt{n}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{z^2}{2}\right) dz$$

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Proof plan:

1) Rely on Fact that convergence of MGFs $\rightsquigarrow$ convergence in CDFs
2) Show that MGFs, in limit, converge on normal MGF

Proposition

Let F_n be a sequence of cumulative distribution functions with corresponding moment generating functions M_n. F be a cdf with the moment generating functions M. If $\lim_{n \to \infty} M_n(t) \to M(t)$ for all t in some interval, then $F_n(x) \rightsquigarrow F(x)$ for all x (when F is continuous).

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Suppose $\lim_{n\to\infty} c_n \to c$, then

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Proposition

Suppose $M(t)$ is a moment generating function of some random variable X. Then $M(0) = 1$.

Proof of Central Limit Theorem

Proof. Suppose $X_1, \ldots, X_n$ are i.i.d. variables with $E[X] = 0$, variance σ_x^2, Moment Generating Function (MGF) $M_x(s)$.

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$M_{S_n} = (M_x(s))^n$ and $M_{Z_n}(s) = \left(M_x \left(\frac{s}{\sigma_x \sqrt{n}} \right) \right)^n$

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Using Taylor's Theorem we can write

$$M_x(s) \quad = \quad M_x(0) + sM_x^{'}(0) + \frac{1}{2}s^2 M_x^{''}(0) + e_s$$

$e_s/s^2 \to 0$ as $s \to 0$.

$$M_x(s) = M_x(0) + sM_x^{'}(0) + \frac{1}{2}s^2 M_x^{''}(0) + e_s$$

Filling in the derivatives of M_x we have

$$M_x(s) = 1 + 0 + \frac{1}{2}s^2\sigma_x^2 + \underbrace{e_s}_{\text{Goes to zero}}$$

Using the fact that $M_{Z_n}(s) = (M_x(\frac{s}{\sigma_X\sqrt{n}}))^n$

$$\begin{aligned} M_{Z_n}(s) &= \left(1 + \frac{1}{2}\left(\frac{s}{\sigma_X\sqrt{n}}\right)^2 \sigma_x^2\right)^n \\ &= \left(1 + \frac{s^2/2}{n}\right)^n \\ \lim_{n\to\infty} M_{Z_n}(s) &= e^{\frac{s^2}{2}} \end{aligned}$$

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Estimators

- Statistics are functions of random variables.
- An *estimator* is statistic that is used to infer the value of an unknown parameter.
- An estimate is a particular realization of the estimator.
- Estimators are written with a $\hat{\theta}$.
- Estimators may provide a single point or an interval.

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Properties of Estimators

- Estimators have distributions, expectations and variances.
- Does the expected value of the estimator equal the true value of the parameter? Bias
- Does the estimator minimize its variance? Efficiency
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Bias

The bias of an estimator is defined to be

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

.

- If $E(\hat{\theta}) > \theta$, the estimator is biased upward.
- If $E(\hat{\theta}) < \theta$, the estimator is biased downward.
- If $E(\hat{\theta}) = \theta$, the estimator is unbiased.

Example of Unbiased Estimator

- Sample mean $E(\bar{X}) - \mu$

$$E[\bar{X}_n] - \mu = 0$$

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$$E[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^{n} E[X_i]$$

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- Uncorrected sample variance: $E[\hat{\sigma}^2] - \sigma^2$.

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&= \sigma^2 - \frac{\sigma^2}{n}
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Efficiency

Given two unbiased estimators, we can compare their sampling error.

$$\sigma^2_{\hat{\theta}} = E[(\hat{\theta} - E(\hat{\theta}))^2]$$

- Estimator $\hat{\theta}$ is more efficient than $\tilde{\theta}$ if $\sigma_{\hat{\theta}} < \sigma_{\tilde{\theta}}$
- The estimator with a smaller sampling error is more efficient.
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- Sample Mean: $\sigma^2_{\bar{X}} = E((\bar{X} - \mu)^2)$

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&= \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}
\end{aligned}$$

$$\sigma^2_{\bar{X}} = \frac{\sigma^2}{n}$$

Example of Efficiency

- 'First Impression' Estimator: $\mathcal{W}(X_1,\ X_2,\ X_3, ...) = X_1$

$$(0.1)$$

- Comparing $\sigma^2_{\mathcal{W}}$ to $\sigma^2_{\bar{X}}$

$$\sigma^2 > \frac{\sigma^2}{n}$$

Example of Efficiency

- 'First Impression' Estimator: $\mathcal{W}(X_1,\ X_2,\ X_3, ...) = X_1$

$$\sigma^2_{\mathcal{W}} = E((X_1 - \mu)^2) \tag{0.1}$$

- Comparing $\sigma^2_{\mathcal{W}}$ to $\sigma^2_{\bar{X}}$

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Example of Efficiency

- 'First Impression' Estimator: $\mathcal{W}(X_1,\ X_2,\ X_3, \ldots) = X_1$

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Mean squared Error

- If we have a biased estimator, we can compare sampling error by evaluating *mean squared error*

$$\text{MSE}(\hat{\theta}) = E(\hat{\theta} - \theta)^2$$

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Design of Studies with a known μ and σ

- How many draws should we get to make sure that our sample mean is close to the truth?
- Suppose that we assume that our data is normal, with a known mean μ and σ^2
- What is

$$P((\bar{X}_n - \mu)/(\sigma/\sqrt{n}) < z|\mu,\ \sigma^2, n)$$

- Fix α, and we can calculate n.

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Design of Studies

- All we need to do is find the smallest n that produces:

$$P((\bar{X}_n - \mu)/(\sigma/\sqrt{n}) < z|\mu,\ \sigma^2, n) \leq \alpha$$

- And solve for n.

Sample size when we know μ and σ

- Imagine that you are running a survey, and have to pick a sample size ahead of time such that

$$P(|\bar{X} - \mu| < \sigma/5) = .95$$

- What sample size would you choose? $\Phi^{-1}(.95) = 1.644854$

$$\Phi^{-1}(.975) = 1.959$$

$$P(\frac{|\bar{X}_1 - \mu|}{\frac{\sigma}{\sqrt{n}}} < 1.96) = .95$$

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- What if you have a random variable with distribution $X \sim N(\mu, 12)$ and you want $|\bar{X} - \mu| < 3$ with a probability of .95.

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Design of studies

- If you are asked, ‘How large a sample size do I need’
- Ask:
- How accurately do you need the answer?
- What level of confidence do you intend to use?
- What is your current estimate?

Finite Sample Properties of Estimators

- If $X_i \sim N(\mu,\ \sigma^2)$

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

- $\bar{X}$ and S^2 are independent:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

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